



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Civil

Course Code & Title: CME365 Renewable Energy Technologies

Name of the Faculty member (s): Dr. S. Narthana

Innovative Practice Description

- **Unit / Topic:** Unit I / Types of energy sources and comparison between conventional and non-conventional sources
- **Course Outcome:** CO 1
- **Unit Outcome:** TLO1
- **Activity Chosen:** One Minute Paper

- **Justification:**

The types of energy sources and differences between the conventional and non-conventional sources are identified using this approach. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of existing energy resources. This technique helps in analyze the energy sources and improves the students' learning with the focused thinking.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	1	1	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 10
	1
Justification for correlation	The students can effectively communicate on engineering activities and make presentations on the given topics

• Images / Screenshot of the practice:

Innovative Teaching Method Execution

Types of energy sources and comparison between conventional and non-conventional sources – One Minute Paper

CPE 365 - Renewable Energy Technologies
 Name: P. Nivedha
 Reg No: 953601103024

1) Classification of Energy Sources

Sources of Energy

conventional source

commercial

eg:
 * coal
 * petroleum
 * electricity

Non-commercial

eg:
 * firewood
 * straw
 * Dried dung

non-conventional source

eg:
 * Bioenergy
 * Solar energy
 * wind energy
 * Tidal energy
 * Energy from urban waste

2) Compare the conventional and non-conventional energy sources

conventional energy sources	non-conventional energy sources
* These sources of energy are not abundant, present in limited quantity. Eg: coal, petroleum, natural gas * They have been in use for a long time * They are called non-renewable sources of energy	* These sources of energy are abundant in nature eg: Solar energy, wind energy, tidal energy, biogas from biomass * They are yet in development phase over the past few years * They are called renewable sources of energy

Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ❖ **Benefit of the practice:**
- ❖ Students can be able to attend the question even the questions are in indirect form.
- ❖ Students can able to explain the concepts in examination without any confusion.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
2. Renewable Energy Sources and Emerging Technologies, by Kothari, Prentice Hall India Learning Private Limited; 2nd edition (1 January 2011), ISBN-10:8120344707.

Signature of Faculty Member

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Innovative Practice Description

Unit / Topic: Unit II / Demonstration of Solar thermal collector and PV System Operation

- **Course Outcome:** CO 2

- **Unit Outcome:** TLO11

- **Activity Chosen:** Field Visit

- **Justification:**

The demonstration of the types of collectors and various applications of solar PV systems are virtually shown to the students through this type of approach. The power rating of the installed solar panels and working of solar water heating systems are explained. This technique helps in understanding and analyzing the various applications of the PV systems virtually and hence improves the students' learning.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

Field visit require students to actively engage in their learning, often by connecting their prior knowledge to new information. When going for such a visit, a student will be understanding the basics and identify the specific topics in solar thermal applications. The students actively participated in this event.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	-	-	-	2	2	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	The students can be able to understand and identify the engineering products

- **Images / Screenshot of the practice:**



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during field visit.
- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The benefits of field visits are a great way for students to make notes on all of the information they receive. It made the students to identify the basic concepts in construction and working principle of solar PV systems involved in the process and make the students to deliver the same for the particular topic.

- ❖ **Challenges faced in implementation:**

1. Time utilization for conducting activity.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, Graw Hill; First edition (10 December 2020), ISBN-10:9390385636.
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Innovative Practice Description

- **Unit / Topic:** Unit III / Wind Energy
- **Course Outcome:** CO 3
- **Unit Outcome:** TLO14
- **Activity Chosen:** Ungraded Quiz
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**

Ungraded quiz innovative practice was conducted for IV year Civil students, after explained the concept of Wind energy and its conversion principles. Quiz activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. The quiz has been created in the Quiz online tool and pass code has been shared with the students.

 - All the students have started the quiz at the same time. At the end of the quiz, the correct answers will be shown to them in the screen.
 - Finally, I explained the correct answer for the quiz questions with proper reasons.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	-	-	-	2	2	1	-	1	-	1	-	-	-

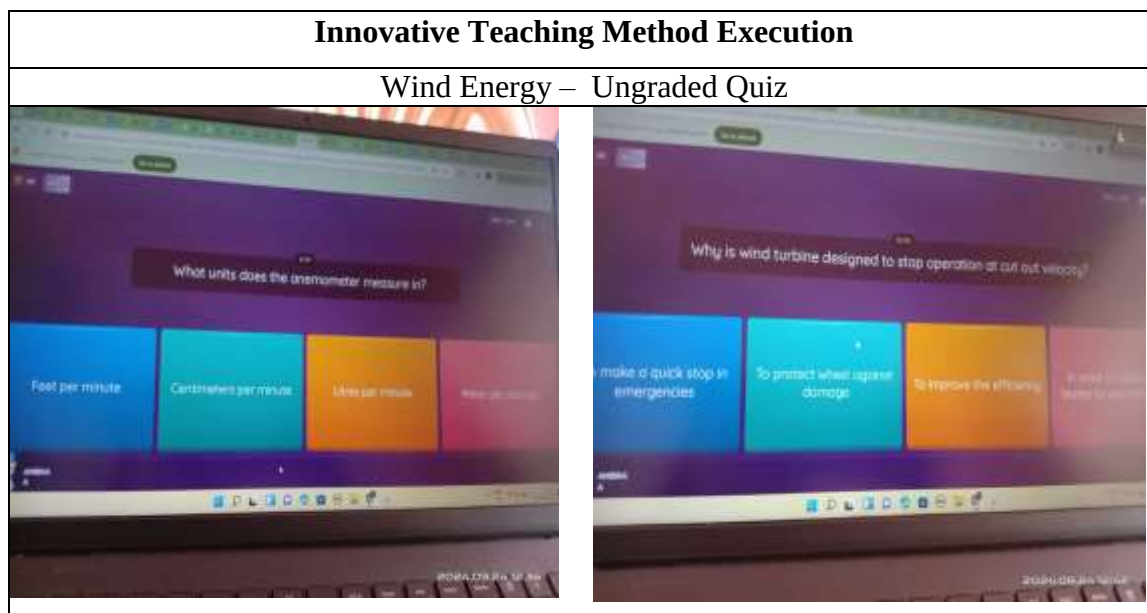
(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO 5	PO 8
	1	1

Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 1	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1
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• **Images / Screenshot of the practice:**



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students understood the important terminologies and design principles present in this topic and improved the ability to relate their answers to the classroom learning.

❖ **Benefit of the practice:** The benefits of quiz are a great way for students to make notes on all of the information they receive. It made the students the general steps involved in the process and make the students to implement the procedure for the particular topic.

❖ **Challenges faced in implementation:**

Normally teachers will give longer explanations in the notes section of the topic. The students are made to answer the multiple-choice questions, fill ups for the topic of wind energy only for 15 mins. But in real scenario, it takes 25 mins.

References:

1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Biomass direct combustion
- **Course Outcome:** CO 4
- **Unit Outcome:** TLO18
- **Activity Chosen:** Flipped Classroom
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
 - ✓ This activity greatly helps the student to know the concept much deeper.
- **Time Allotted for the Activity:** 25 minutes
- **Details of the Implementation:**

The team of 2 students were asked to do literature survey about the given topic. They should include at least 5 Reference material about the given topic for which the material is already shared. Students in a team were asked to present their topics. Queries will be asked by the students to the presenting team.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	-	-	-	2	2	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO 10
	1
Justification for correlation	The students can effectively communicate on engineering activities and make presentations on the given topics

- Images / Screenshot of the practice:



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

The students enjoyed this Flipped classroom Activity. After the class many students requested me to insist other Course faculty members to conduct this activity. I have taken this feedback to by Head of the Department

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students actively participated in the flipped classroom activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process

- ❖ **Challenges faced in implementation:**

Still few students were hesitant to convey their points and queries. Few students have not taken this activity seriously and remain quite in the class

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1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
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Innovative Practice Description

- **Unit / Topic:** Unit V / Environmental impact of ocean and geothermal energy
- **Course Outcome:** CO 5
- **Unit Outcome:** TLO27
- **Activity Chosen:** One Minute Paper
- **Justification:**

The environmental impact of other energy sources is identified using this approach. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of other energy resources apart from solar and wind energy. This technique helps to analyze the environmental sustainability and its effects with the usage of ocean and geothermal energy sources.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	-	-	-	2	2	1	-	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 10
	1
Justification for correlation	The students can effectively communicate on engineering activities and make presentations on the given topics

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Environmental impact of ocean and geothermal energy – One Minute Paper
Environmental problems from Geothermal Energy. * The geothermal fluids have dissolved solids entrained solid particles, non condensable gases & sand. These solids & non condensable gases create following environmental problems. Emissions :- * The emissions of H_2S, CO_2 & other toxic gases into the atmosphere during operation of the plant is a source of pollution to the environment. * H_2S gas has annoying smell geothermal plant may emit about 520 kg of H_2S gas per day in the atmosphere. Effluents :- * The effluents from geothermal fluids during conversion are highly mineralised. These effluents can pollute the water streams or underground water source if disposed of without treatment. * certain effluents contain boron, fluorines, mercury & arsenic compounds which are extremely harmful to animal & plant life.

Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ❖ **Benefit of the practice:**
- ❖ Students can be able to attend the question even the questions are in indirect form.
- ❖ Students can able to explain the concepts in examination without any confusion.
- ❖ **Challenges faced in implementation:**
- 1. Time utilization for conducting activity.

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1. Fundamentals and Applications of Renewable Energy | Indian Edition, by Mehmet Kanoglu, Yunus A. Cengel, John M. Cimbala, cGraw Hill; First edition (10 December 2020), ISBN-10:9390385636.
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